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Lower Permian paleomagnetism of the Tarim block, northwestern China

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We collected 262 samples at 39 sites in three thick Lower Permian sections along the northwestern margin of the Tarim block. The Lower Permian sections are basalts interbedded with limestones or sandstones, or shallow marine limestones and sandstones alone, all with biostratigraphic age control. Alternating field (AF) and thermal demagnetization experiments remove small secondary components and reveal characteristic magnetizations. Reversed directions (consistent with the Kiaman Reversed Interval), similar directions between basalts, sandstones and limestones, a positive fold test, and apparent polar wander and polarity changes recorded in underlying Devonian sediments all attest to the primary nature of the remanence. Two paleomagnetic poles, one from the limestone section ($\lambda = 65.1°$ N, $\phi = 162.5°$ E, $K = 128$, $A_{95} = 4.1°$, $n = 11$ sites) and the other from the two basalt sections ($\lambda = 54.5°$ N, $\phi = 172.3°$ E, $K = 66$, $A_{95} = 4.0°$, $n = 21$ sites), differ primarily in colatitude ($8.1 \pm 6.5°$) suggesting a difference in magnetization age. Permian paleomagnetic poles from Tarim, Siberia, North China and South China all differ significantly suggesting substantial post-Permian relative motions.

1. Introduction

China is one of the most complicated geologic terranes in the world. The Phanerozoic tectonic development of China has two stages: a Paleozoic stage of convergence and a Mesozoic stage of collision and accretion [1–3]. Gradual rapprochement of the Siberian, North China, South China and Tarim blocks closed intervening Paleozoic ocean basins and generated major Late Paleozoic and Mesozoic accretionary zones. To understand detailed motions and interactions of the major continental blocks, reliable paleomagnetic data are helpful; the past emphasis in paleomagnetic data collection has been in eastern China (both the North China and South China blocks [4–9]), however, few data exist from the Tarim block, the major continental mass in western China [10–15]. Of the three great blocks that compose the tectonic framework of China, the Tarim block is the most inboard (see Fig. 1). Therefore it must have arrived at its present position at least as early as any of the more outboard blocks and terranes. Just as the apparent polar wander for the North American craton provides a reference for assessing the motion of terranes of the Cordillera, Tarim paleomagnetic data will provide a reference frame for estimating motions of the terranes and blocks that lie between it and the Pacific and Indian basins.

We have undertaken a paleomagnetic study of Lower Permian sequences to measure the translation and/or rotation of Tarim, and to locate its Late Paleozoic paleogeographic position relative to other tectonic units in Asia.

2. Geologic setting

The Archean to Proterozoic basement of the Tarim block is covered by Late Proterozoic and Phanerozoic strata [16,17]. The interior of the block is mostly blanketed by sand dunes of the Taklimakan Desert. Late Precambrian (Sinian) clastics, volcanics and tillites, Paleozoic carbonates

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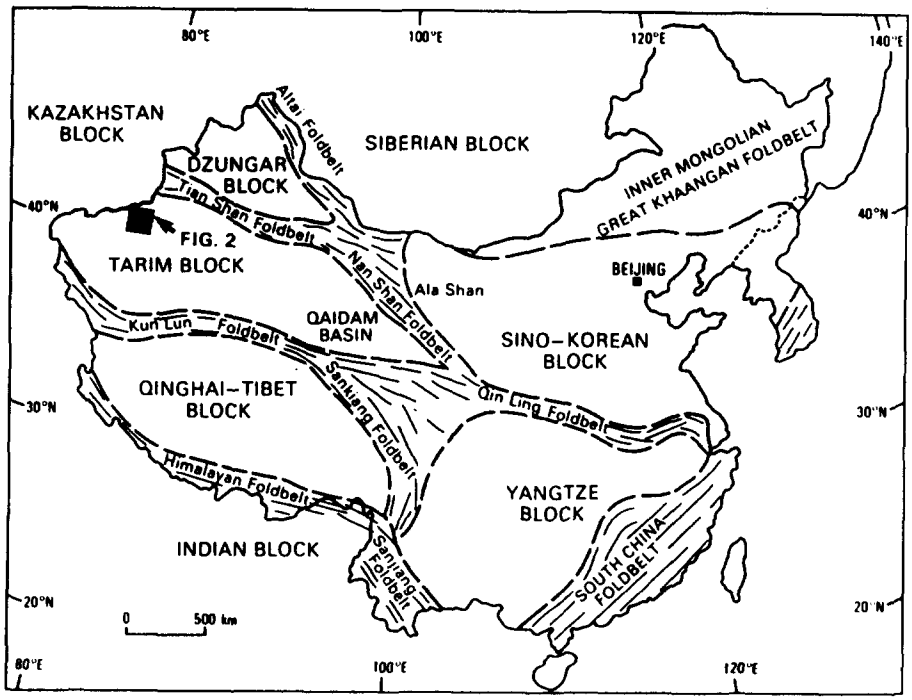


Fig. 1. Generalized tectonic map of China after McElhinny and co-workers [6,29].

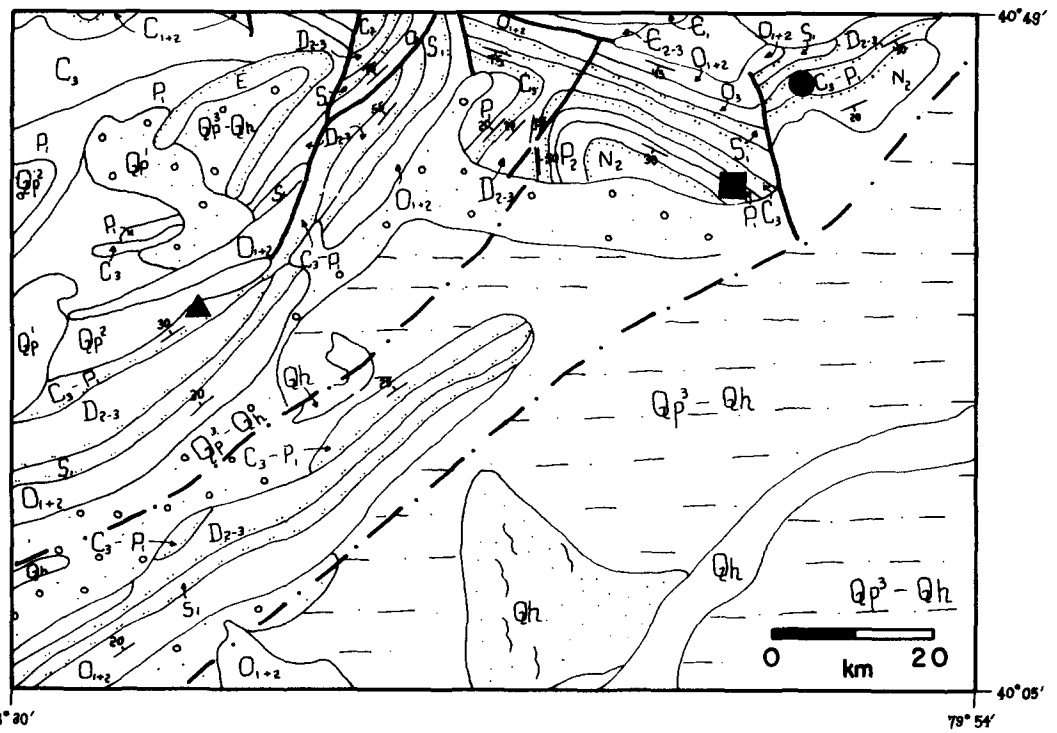


Fig. 2. Generalized geological map of the Aksu-Keping region of Tarim following the Xinjiang Geology Bureau [30]. The Subasi, Yingan and Camp 66 locations are labeled with triangle, square and circle symbols, respectively. The subscript 1 indicates the lower part of the geologic time period, the subscript 2 middle or upper, and the subscript 3 upper. For example, P_1 labels Lower Permian strata, P_2 Upper Permian strata and C_3 Upper Carboniferous strata. Bold lines, solid and dashed, mark actual and inferred fault traces, respectively. Formation contacts drawn together with a continuous dotted line indicate unconformities, while those together with double-dot dotted lines mark pseudo-conformities.

and marine clastics, and Mesozoic to Cenozoic continental clastics, red beds and evaporites are well exposed at the periphery.

2.1. Regional stratigraphy and structure

Our sampling was concentrated in the Aksu-Keping area on the northwestern margin of the Tarim block (see Fig. 2). The structures sampled are gently dipping sequences spanning Cambrian to Lower Permian time continuously, except for a few disconformities. The stratigraphic column is repeated by a series of southwest–northeast striking, apparently southeast-vergent thrust faults. The lithologies are suggestive of a stable platform depositional environment: Cambrian and Ordovician units are carbonates with shallow water marine fauna (primarily trilobites). Silurian deposits are thin where present, and consist of green marine, argillaceous sandstone, followed by spectacular exposures of Devonian fluvial/alluvial red sandstone up to 2000 m in stratigraphic thickness. A major disconformity separates the Devonian rocks from Upper Carboniferous shallow water marine limestones abundant with coral, and interbedded mature, well-sorted sandstone layers [18]. Completing the Paleozoic stratigraphy, Lower Permian sections are either shallow water marine limestones and sandstones, or basalt flows with interbedded limestone or coarse fluvial sandstone layers and coal (see Fig. 3). The folding and faulting which produced these fault blocks and detached them from the basement may represent a Tertiary compressional event resulting from the convergence between the Indian and Eurasian plates [19].

2.2. Sampling localities

We collected Lower Permian samples at three stratigraphic localities: near the villages of Subasi (40.50° N, 78.78° E), Yingan (40.67° N, 79.56° E) and Camp 66 (40.73° N, 79.70° E). Lower Permian stratigraphy at the Subasi locality consists of the Balikelike Formation, 150 m of shallow marine, organic-rich limestone, overlain by the Kalundal

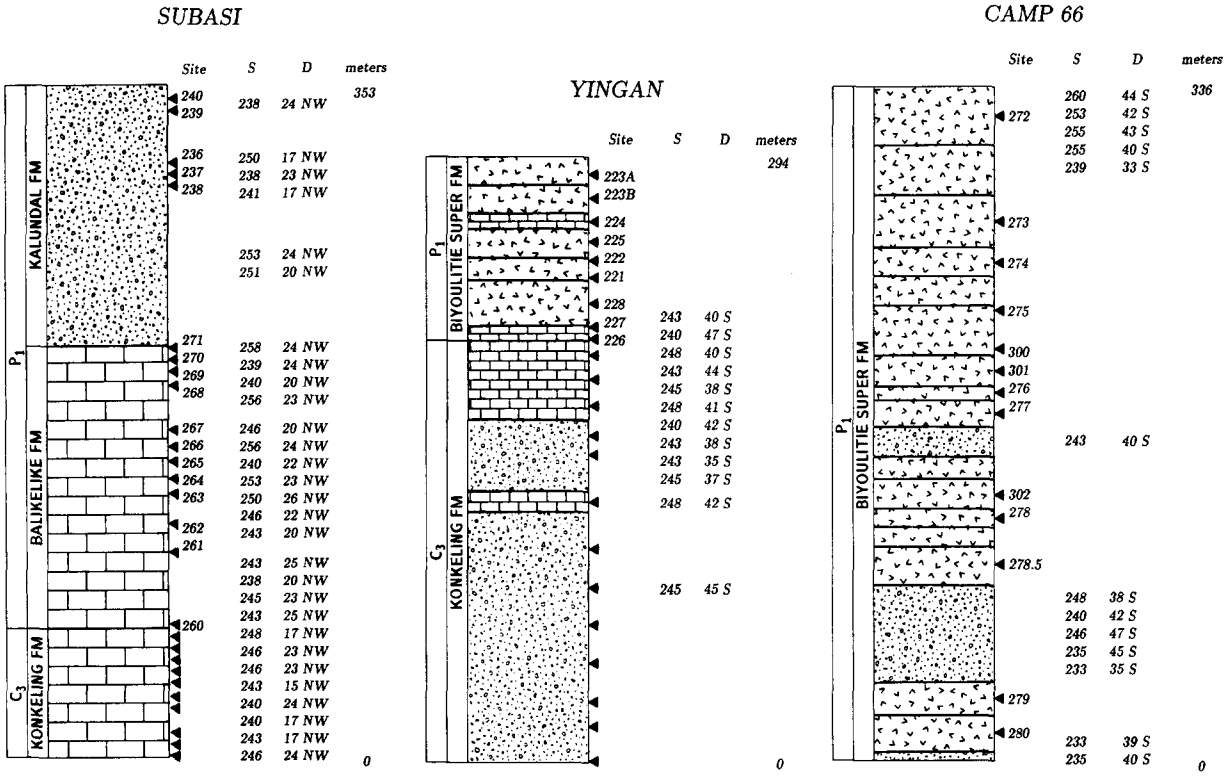


Fig. 3. Stratigraphic sections showing the distribution of sampling sites and strike and dip measurements for each of the three locations. Note change in scale from section to section.

Formation, an additional 150 m of gray to red, medium-grained sandstone (see Figs. 2 and 3). Exposures were fresh and accessible with the entire unit dipping monoclinally toward the northwest. Poles to bedding planes for 29 strike and dip measurements (individual readings all listed in Fig. 3) taken throughout the sequence were averaged to give a final strike of 245.5° and a dip of 21.5° NW ($k = 471$, $\alpha_{95} = 1.2°$). Twelve sites were evenly distributed within the limestone unit, with an additional five toward the top of the sandstone. On average, six samples were taken from each site.

A second, southeastward-dipping section consisting of 49 m of basalt flows with interbedded limestones (all part of the Lower Permian Biyoulitie Super Formation) was sampled at Yingan 70 km northeast of Subasi (see Figs. 2 and 3). Nine sites with six samples per site were collected. Three sites were located within thin limestone layers and the remaining six in six distinct volcanic cooling units. The interbedded limestones are replete with algal debris and indicate a shallow marine or tidal flat depositional environment corresponding to a hiatus in volcanic activity. An average of twelve poles to bedding planes from sites within the limestones and in the underlying Carboniferous unit gives an overall strike and dip of 64.1° and 40.7° SE ($k = 420$, $\alpha_{95} = 2.1°$).

The third section was sampled at Camp 66, an additional 15 km to the northeast of Yingan (see Figs. 2 and 3). Here, in the more continental end member of the Biyoulitie Super Formation, basalt flows are interbedded with southeast-dipping sandstones for a total of 336 m of section. Only the basalts were sampled, and thirteen sites were placed in twelve distinct cooling units. Eight samples were collected at each site. The sandstones provide good attitude control. An average of thirteen poles to bedding planes establishes a section strike and dip of 64.4° and 40.3° SE ($k = 127$, $\alpha_{95} = 3.7°$).

2.3. Biostratigraphy

Paleontologic analyses (courtesy of Chevron Overseas Petroleum Inc.) of three limestone samples from the Balikelike Formation at Subasi reveal diagnostic Permian fauna. A sample from Site 261 is a packstone with a microsparite matrix. Grains are composed of a diversity of encrusting foraminifers, including *Colaniella, Pachyphloia, Paraglobivalvulina* and *Nankinella*. Other fossils present are gastropods and algal fragments (*Macroporella* and *Velomorpha*). Fossil content and lithology are indicative of a shallow lagoonal depositional environment. Higher in the section, a sample from Site 265 is characterized as a dark brown, very silty and organic-rich wackestone with scattered, disarticulated and broken shell fragments. Many of the fragments are unidentifiable, however, some consist of pieces of brachiopods, bryozoans, thin-shelled pelecypods, crinoid columns, cephalopods, fish bones and miliolid foraminifers. One of the miliolids is *Agathammina*, ranging in age from the Namurian through the Zechstein of Northwest Europe. Petroleum stains are common, though total organic carbon content is less than 1% by weight. A sample from the top of the Balikelike exposure, from Site 270, is a medium to light brownish gray wackestone to packstone. Common productid brachiopod spines are present, as well as some algae (*Macroporella*), cephalopod fragments, rare fenestellid bryozoans, encrusting miliolid forminifers and ostracodes.

All of the identified fossil forms could range through the Maokouan Stage into the overlying Wujiapingian Stage of China (from the Weissliegendes into the Buntsandstein of Northwest Europe [16]), however, the most diagnostic fauna point to an age similar to that of the lower part of the Maokouan of eastern China. These age determinations agree well with previous Chinese age assignments for the Balikelike [16]. Ages for the volcanic sequences exposed at both Yingan and Camp 66 are much less well known. No isotopic ages are available, but the interbedded sedimentary layers do yield some Permian fossil material: the bivalve *Palaeanodonta* in the limestones at Yingan and the plant *Pecopteris* in the sandstones at Camp 66. Relative ages between the Balikelike Formation at Subasi and the two basalt sections are uncertain, although all three sections overlie the same Upper Carboniferous Konkeling Formation.

3. Paleomagnetism

Two hundred and sixty-two samples from the 39 sites distributed throughout the three Lower Permian sections were collected during August and September of 1986. The azimuthal orientation

of each core consisted of a magnetic reading for all samples, and a sun azimuth for 188 samples. The absence of local magnetic field gradients at sites without sun readings was quickly verified in the field by reference to a distant point. Two specimens were cut from 53 of the longer cores, an A specimen at the bottom of the core and a B specimen closer to the original outcrop surface. Magnetic measurement and demagnetization procedures were conducted at Stanford University using an automated cryogenic magnetometer with integrated, alternating field (AF) demagnetization coils, and a Schonstedt TSD-1 oven. For specimens subjected to thermal demagnetization, bulk magnetic susceptibilities as a function of heating step were monitored with a Bartington magnetic susceptibility meter, model M.S.2.

3.1. Pilot study

To provide a preliminary assessment of the magnetization, two A specimens, a Subasi limestone and a Yingan basalt, were cleaned magneti-

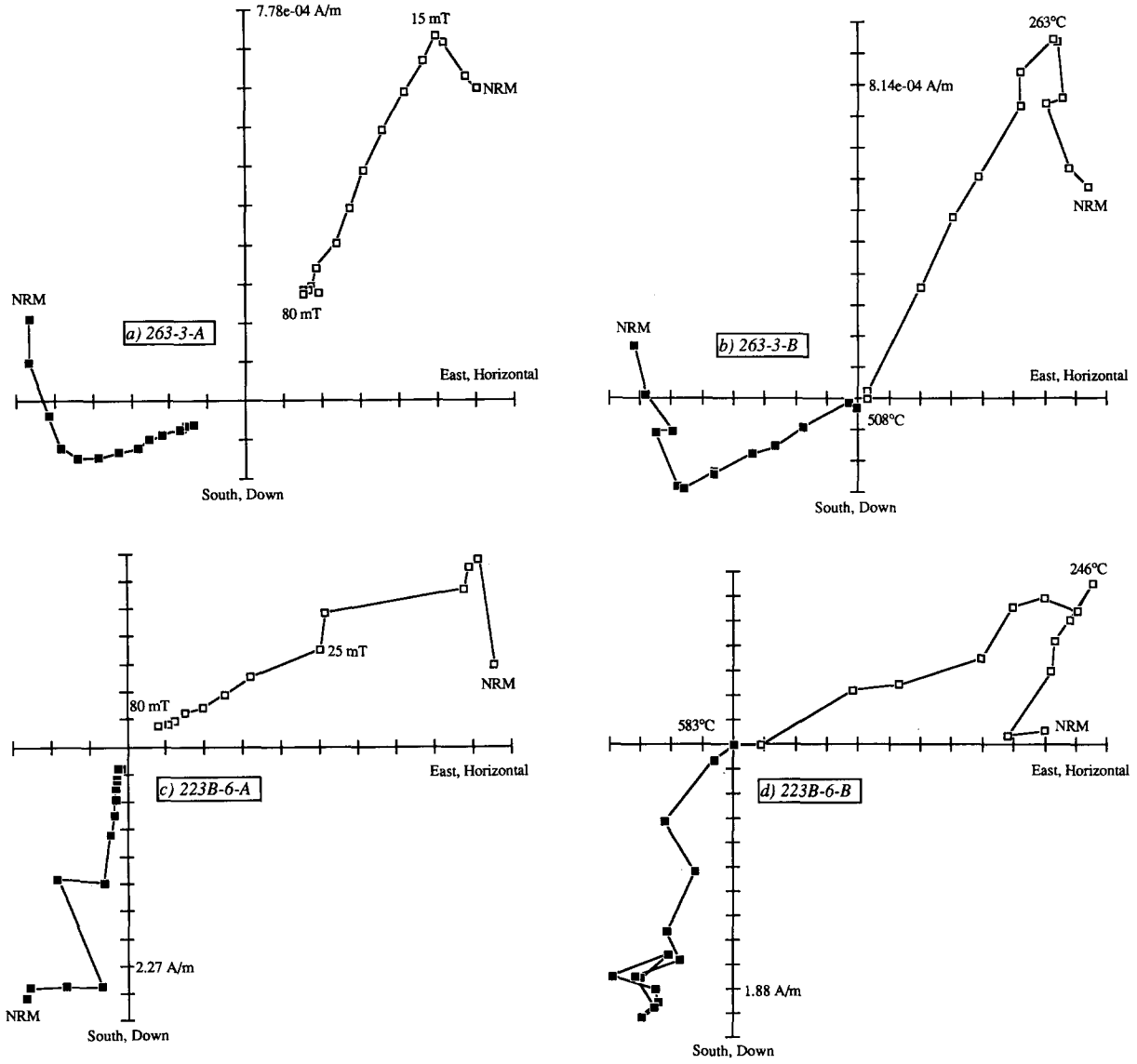


Fig. 4. Orthogonal vector plots in in situ coordinates of AF and thermal pilot demagnetization. Closed symbols are projections onto the horizontal plane and open symbols are projections onto a vertical plane. Data from two Subasi limestone specimens are plotted in parts (a) and (b), and from two Yingan basalt specimens in parts (c) and (d).

cally while the two corresponding B specimens were cleaned thermally. The A specimens were demagnetized in alternating fields with peak values ranging from 5 to 80 mT. In parallel experiments the B specimens were heated in the Schonstedt oven in which the residual field in the cooling chamber was 3 nT. To remove any soft components of magnetization acquired while transferring the specimen into the magnetometer from the oven, the specimens were subjected to an alternating field of 5 mT within the field-free space of the magnetometer.

In both pairs of specimens a component of magnetization was present with coercivities (H_C) in excess of 80 mT in the A specimens, and with blocking temperatures (T_B) in the range of 250–450 °C for the limestone B specimen and in the range of 250–580 °C for the basalt B specimen (see samples 263-3 and 223-6 in Fig. 4). A low H_C/T_B component of magnetization present in these four specimens was removed in alternating fields of 15 mT and at temperatures of 250 °C or less. For this pair of limestone specimens, the high H_C component found after AF demagnetization differed from the high T_B component found as a result of the thermal treatment. The coercivity spectra of the low H_C and high H_C components of magnetization appeared to overlap, while their blocking temperature spectra tended to be more distinct. For the pair of basalt specimens, the magnetic directions found by AF cleaning were much less scattered than those found by thermal cleaning.

3.2. Subasi section

All of the remaining A specimens from the twelve sites within the limestone Balikelike Formation at Subasi, Sites 260–271, were thermally demagnetized. Each specimen was heated 13 times in 50 °C increments from 50 ° to 250 °C and then in 25 °C increments to a final temperature of 450 °C. The thermal demagnetization behavior of specimens from Sites 261–271 was very similar to that of the limestone pilot specimen 263-3-B (see Figs. 4b and 5a). Each demagnetization path consisted of two components of magnetization with discrete blocking temperature spectra, a low T_B normal component and a higher T_B reversed component. Intensities of natural remanent magnetization (NRM) were on the order of $5–10^{-4}$ A/m, and grew by a factor of 1.5, to about 8×10^{-4} A/m, in response to the removal of the normal component during heating to 250 °C. The reversed component of magnetization was isolated during heating at 275 °C, and decayed univectorially toward zero with additional heating to 450 °C. Bulk magnetic susceptibilities dramatically increased during heating steps greater than 450 °C, indicating chemical alteration of the specimens. Final intensities of magnetization averaged 8×10^{-5} A/m, an order of magnitude less than intensities after removal of the normal component. Specimens from Site 260 exhibited anomalous thermal demagnetization behavior in that each failed to give a coherent high T_B direction (see Fig. 5b). In each specimen an NRM similar in direction to the present earth's field was removed on heating to 150 °C or less. No useful component of magnetization could be identified. This directional aberrance correlates well with a unique lithology. These pale limestones from Site 260, the lowest site in the stratigraphic sequence at Subasi, lack the dark organic-rich muds, apparently the essential ingredient for magnetic stability within the Balikelike Formation.

"A" specimens from the five sites within the sandstone Kalundal Formation at Subasi, Sites 236–240, were also thermally demagnetized. Each specimen was heated 16 times in 50 °C increments from 50 ° to 500 °C and then in 25 °C increments to a final temperature of 650 °C. Two different types of thermal demagnetization behavior were observed among samples from the five sites:

Type 1. Specimens from Sites 236, 237 and 240 were heavily overprinted by a component of magnetization similar in direction to the present earth's field. Intensities of NRM were on the order of 8×10^{-3} A/m and fell by an order of magnitude in response to the removal of the substantial normal component during heating to 300 °C. A higher T_B reversed component of magnetization was present, and in almost all cases a direct observation was made (see Fig. 5c). Successive demagnetizations typically moved the vector observed along a great circle away from the present earth's field direction to the underlying characteristic direction.

Type 2. The thermal demagnetization behavior of specimens from Sites 238 and 239 was different in

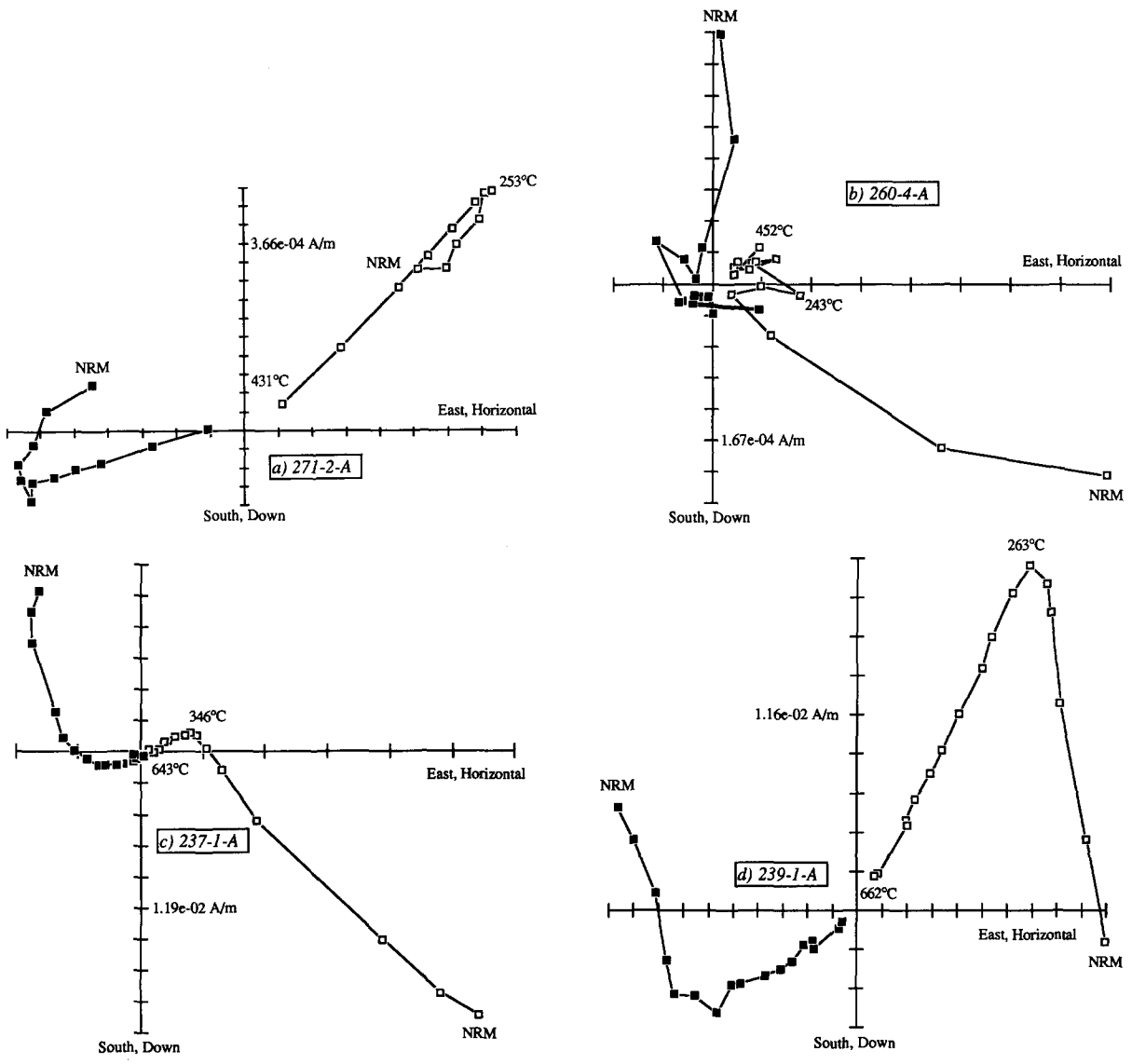


Fig. 5. Representative orthogonal vector plots in in situ coordinates of thermal demagnetization. Closed symbols are projections onto the horizontal plane and open symbols are projections onto a vertical plane. Data from two Subasi Balikelike Formation limestone specimens are plotted in parts (a) and (b), and from two Subasi Kalundal Formation sandstone specimens in parts (c) and (d).

that the high T_B reversed component of magnetization was as large as the low T_B normal component. NRM intensities on the order of 1×10^{-2} A/m increased 1.5 times of 1.5×10^{-2} A/m as the normal component of magnetization unblocked with heating to 250 °C. The reversed component was well defined as it decayed univectorially to zero further heating to 650 °C (see Fig. 5d). The two components of magnetization were approximately equal, each 1×10^{-2} A/m in magnitude.

3.3. Yingan section

The basalt and interbedded limestones at Yingan gave stable and consistent paleomagnetic results, with both AF and thermal demagnetization techniques resolving two components of magnetization. All A specimens from the six basalt sites, Sites 221, 222, 223A, 223B, 225 and 228, were AF demagnetized. Each specimen was subjected to alternating fields with peak values ranging from 5 mT to 80 mT in 5 mT increments. The AF demagnetization behavior of specimens from

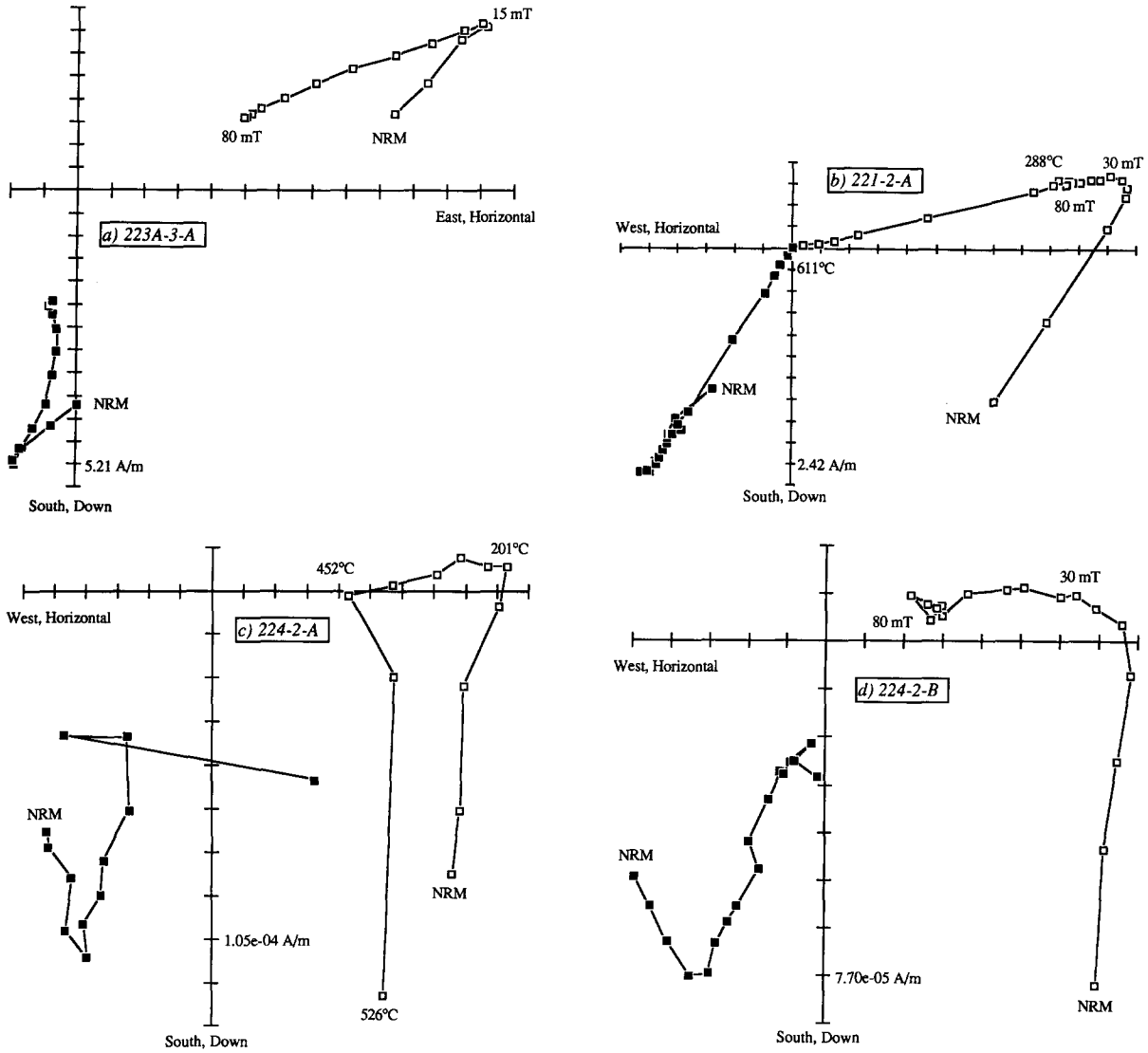


Fig. 6. Representative orthogonal vector plot in in situ coordinates of AF and thermal demagnetization. Closed symbols are projections onto the horizontal plane and open symbols are projections onto a vertical plane. Data from two Yingan basalt specimens are plotted in parts (a) and (b), and from two Yingan limestone specimens in parts (c) and (d).

the six basalt sites was similar to that of the basalt pilot specimen 223B-6-A (see Figs. 4c and 6a). Each specimen contained two components of magnetization: a low H_C normal component and a high H_C reversed component, both with a discrete coercivity spectrum. Intensities of NRM were on the order of 2.9 A/m and grew slightly to 3.1 A/m as the normal component was removed in fields to 20 mT. The reversed component of magnetization was isolated during demagnetization in fields greater than 20 mT, and mean destructive fields ranged from 40 mT to above 80 mT.

For specimens with mean destructive fields exceeding 80 mT, those from Sites 221, 222, 225 and 228, additional thermal demagnetization steps were performed. After AF demagnetization in fields to 80 mT, each of the eight specimens from Site 221 was heated nine times from 300° to 400°C in 25°C increments and then to 600°C in 50°C increments. As before, any low H_C components of

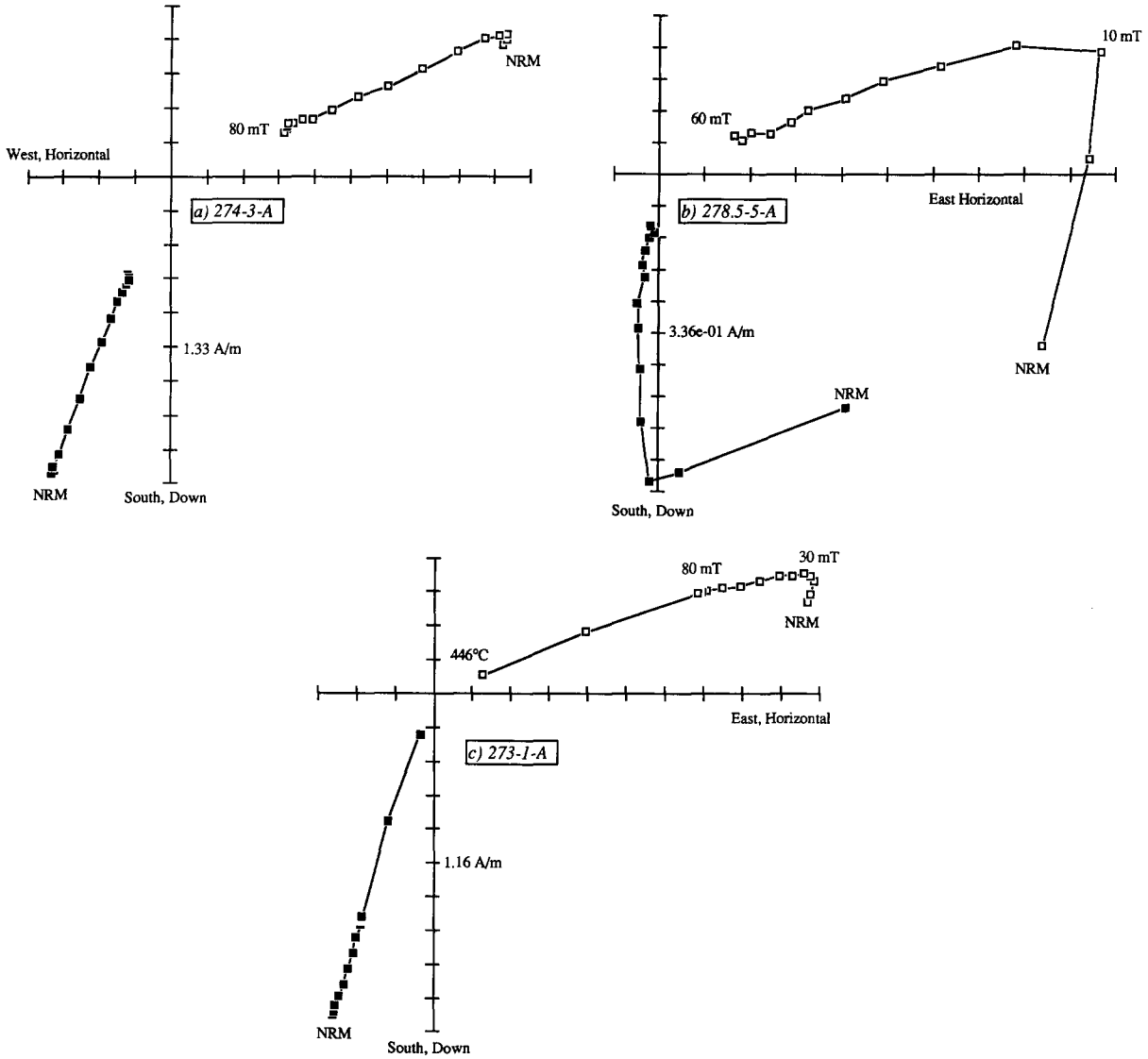


Fig. 7. Representative orthogonal vector plots in in situ coordinates of AF and thermal demagnetization. Closed symbols are projections onto the horizontal plane and open symbols are projections onto a vertical plane. Data from three Camp 66 basalt specimens are plotted in parts (a), (b) and (c).

pole (VGP) positions were calculated. Finally VGP positions were combined to yield provisional paleomagnetic pole positions for each formation.

4.1. Calculation of mean directions

The multivariate technique of principal component analysis described by Kirschvink was used to find the best least-squares fit along the demagnetization path for each specimen.

Subasi section. On average, seven magnetic vectors obtained by thermal demagnetization between 275° and 425° C were used to define the reversed characteristic direction for each specimen from the Balikelike limestone Formation sites, Sites 261–271. For the Kalundal sandstone Formation, Sites 236–240, the reversed characteristic direction for each specimen was determined on average by eight magnetic vectors defined by thermal demag-

netization between 300° and 575°C. Reversed site mean directions of magnetization along with formation means and VGP positions are listed in Table 1.

Directions of the low T_B normal magnetization were well defined on average by six magnetic vectors (those obtained from the NRM and five thermal demagnetization steps to 250°C) for each specimen from all sites, Sites 261–271 and Sites 236–240. Average magnetization directions for the low T_B component for both the Balikelike Formation ($D = 11.4°$, $I = 55.1°$, $k = 27$, $\alpha_{95} = 8.9°$, $n = 11$ sites) and the Kalundal Formation ($D = 355.4°$, $I = 65.9°$, $k = 41$, $\alpha_{95} = 12.1°$, $n = 5$ sites) are indistinguishable from either the present-day field direction ($D = 2.9°$, $I = 59.6°$) or the dipole field direction ($D = 0.0°$, $I = 59.7°$) at Subasi.

We have decided to exclude the reversed magnetizations isolated from the Kalundal sandstone specimens, Sites 236–240, for two reasons. First, the overall inclination ($\bar{I} = -48.9°$) for the Kalundal sandstone is 9° shallower than the inclination ($\bar{I} = -57.7°$) for the underlying Balikelike limestone. This shallow inclination could reflect apparent polar wander, a substantial difference in magnetization age. A more likely explanation, however, is a systematic depositional inclination error in the sandstones associated with a depositional remanent magnetization (DRM) [21].

The change in direction of magnetization with change in lithology, along with apparent polar wander and polarity changes recorded in Devonian sediments beneath the Kalundal [14], precludes the possibility of a post-depositional regional overprint for the sequence found at Subasi.

Second, directions of reversed magnetizations from Sites 236, 237 and 240 are not well determined due to substantial (often 20 times greater) relative size of the low T_B normal overprint. Results from just two sites (Sites 238 and 239) can hardly be representative of the Kalundal Formation as a whole.

Assuming that the VGP positions are drawn from a Fisherian distribution, the limestone portion of the Subasi section establishes a Lower Permian paleomagnetic pole at $\lambda = 65.1°$N, $\phi = 162.5°$E, $K = 128$, $A_{95} = 4.1°$, $n = 11$ sites.

Yingan section. Principal component analysis identified reversed and normal components of magnetization for each Yingan basalt and limestone specimen. On average seven magnetic vectors resulting from AF demagnetization in fields with peak values ranging from 30 to 60 mT for specimens from Sites 223A and 223B, and nine magnetic vectors obtained by both the AF demagnetization and two additional thermal demagnetization steps at 375°C and 450°C for specimens from Sites 221, 222, 225 and 228 were used in characterizing the reversed direction of magnetization in the basalt specimens. Seven magnetic vectors measured after thermal demagnetization between 275° and 425°C determined the reversed direction of magnetization in the limestone specimens from Sites 224 and 226. Reversed site mean directions of magnetization along with a section mean and VGP positions are given in Table 1.

AF demagnetization steps at 5 mT, 10 mT and 15 mT, plus the NRM direction for most of the basalt specimens, and first subtracted vectors for the remaining, defined the normal overprint component of magnetization. The NRM direction and five thermal demagnetization steps to 250°C were used for the twelve limestone specimens. The mean direction for the normal components of magnetization for the Yingan section is steep ($D = 41.6°$, $I = 77.6°$, $k = 30$, $\alpha_{95} = 11.3°$, $n = 7$ sites) but similar to both the present-day field direction ($D = 2.9°$, $I = 59.8°$) and the dipole field direction ($D = 0.0°$, $I = 59.8°$).

The paleomagnetic pole calculated from the Yingan site mean reversed directions is $\lambda = 51.0°$N, $\phi = 174.3°$E, $K = 61$, $A_{95} = 6.8°$, $n = 8$ sites.

Camp 66 section. Nine magnetic vectors on average obtained by AF demagnetization on the Camp 66 basalts in field with peak values ranging from 20 mT to 60 mT for specimens from Sites 274, 275, 276, 278, 278.5, 279 and 302, and eleven magnetic vectors resulting from AF demagnetization and subsequent thermal demagnetization at 375°C and 450°C for the rest defined the high H_C reversed component of magnetization. Reversed site mean directions of magnetization along with a section mean and VGP positions are tabulated in Table 1.

TABLE 1

Compilation of paleomagnetic data

Site	N	D	I	D_c	I_c	R	k (K)	$\alpha_{95}(A_{95})$	VGP Lat. N	VGP Long. E
Subasi (lat. = 40.50° N, long. = 78.78° E)										
240	5	248.9	−56.6	218.9	−52.1	4.918	49	11.1	58.1	170.4
239	6	248.6	−54.8	220.3	−50.5	5.987	379	3.5	56.4	171.8
236	6	254.9	−53.0	226.8	−51.2	5.847	33	11.9	51.7	166.6
238	6	219.5	−58.6	197.2	−45.3	5.909	55	9.1	70.2	207.4
237	6	251.8	−43.5	231.6	−42.0	5.863	37	11.2	44.2	173.2
Mean	5	245.8	−53.9	219.0	−48.9	4.939	66	9.5		
271	6	250.5	−61.3	214.9	−56.3	5.984	318	3.8	62.7	165.1
270	5	247.8	−57.7	217.0	−52.6	4.994	703	2.9	59.7	170.8
269	6	243.8	−65.0	206.4	−57.0	5.982	273	4.1	69.4	168.3
268	6	273.0	−62.1	229.0	−64.4	5.979	242	4.3	54.5	142.9
267	7	269.5	−65.1	221.3	−65.1	6.959	147	5.0	59.9	142.3
266	6	246.7	−64.4	208.8	−57.4	5.993	765	2.4	67.7	165.8
265	7	237.6	−61.5	206.5	−52.5	6.929	84	6.6	67.7	179.6
264	6	243.1	−61.7	209.7	−54.3	5.986	358	3.5	65.9	172.8
263	6	245.5	−64.6	207.8	−57.2	5.995	975	2.2	68.3	166.9
262	6	251.1	−67.0	207.7	−60.5	5.949	99	6.8	69.2	157.0
261	6	246.2	−62.8	210.4	−56.1	5.987	388	3.4	66.0	168.2
260										
Mean	11	250.2	−63.4	212.2	−57.7	10.953	214	3.1		
Pole	11					10.922	128	4.1	65.1	162.5
Yingan (lat. = 40.67° N, long. = 79.56° E)										
223B	4	182.6	−33.5	217.9	−63.7	3.992	388	4.7	62.2	147.2
223A	4	200.3	−30.6	233.2	−50.8	3.993	421	4.5	46.7	164.3
224	6	195.9	−16.0	214.5	−42.6	5.772	22	14.6	57.1	188.0
225	7	207.9	−20.8	230.2	−39.0	6.981	309	3.4	44.0	177.7
222	7	204.8	−19.2	226.0	−39.8	6.994	1013	1.9	47.5	180.4
221	8	206.8	−17.4	226.3	−37.1	7.987	520	2.4	46.1	182.2
228	6	198.9	−29.2	230.3	−50.7	5.976	211	4.6	48.9	166.1
227										
226	5	201.0	−18.5	221.8	−41.5	4.918	49	11.1	51.4	182.2
Mean	8	200.1	−23.3	225.2	−45.8	7.895	67	6.8		
Pole	8					7.895	67	6.8	51.0	174.3
Camp 66 (lat. = 40.73° N, long. = 79.70° E)										
272	8	205.6	−29.9	236.7	−47.0	7.959	173	4.2	42.5	166.4
273	7	201.2	−23.3	225.9	−45.1	6.979	281	3.6	49.8	175.6
274	7	200.9	−23.7	226.0	−45.6	6.974	227	4.0	49.9	175.0
275	8	197.6	−24.8	223.8	−48.4	7.968	216	3.8	52.8	173.3
300	8	199.8	−28.4	229.8	−49.7	7.984	448	2.6	48.8	167.8
301	8	197.3	−28.1	227.0	−51.0	7.909	77	6.4	51.5	167.9
276	6	194.1	−26.2	221.6	−51.6	5.990	480	3.1	55.8	170.6
277	8	196.7	−30.6	229.3	−53.1	7.978	325	3.1	50.6	163.7
302	7	189.6	−20.1	210.6	−49.3	6.978	270	3.7	63.2	183.1
278	8	189.8	−20.0	210.7	−49.1	7.887	62	7.1	63.0	183.3
278.5	8	184.4	−25.7	209.5	−56.7	7.945	128	4.9	66.9	168.8
279	7	185.7	−22.2	207.7	−53.1	6.978	270	3.7	67.0	178.6
280	8	178.3	−30.8	206.8	−64.0	7.985	460	2.6	69.9	146.6
Mean	13	193.9	−25.9	221.0	−51.4	12.885	104	4.1		
Pole	13					12.825	69	5.0	56.5	170.9

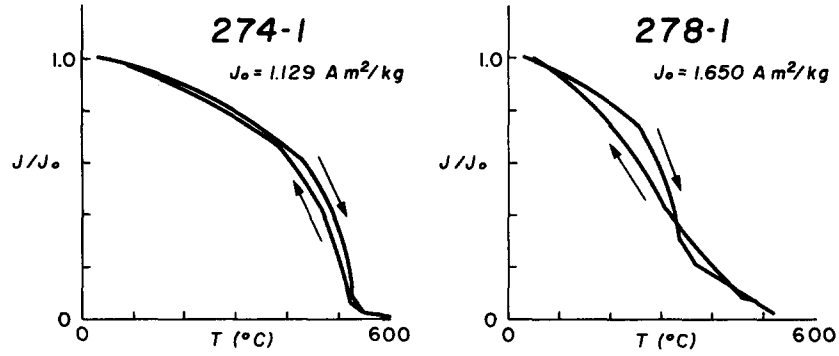


Fig. 8. Normalized saturation magnetization versus temperature analyses for two Camp 66 basalt samples showing the 150° range in Curie temperatures. Analyses all done under a vacuum.

First subtracted vectors establish the mean normal component of magnetization ($D = 43.1°$, $I = 79.6°$, $k = 7$, $\alpha_{95} = 16.9°$, $n = 13$ sites) for the Camp 66 section, a direction similar to either the present-day field direction ($D = 2.9°$, $I = 59.8°$) or the dipole field direction ($D = 0.0°$, $I = 59.9°$).

A paleomagnetic pole for the Camp 66 section is $\lambda = 56.5°$ N, $\phi = 170.9°$ E, $K = 69$, $A_{95} = 5.0°$, $n = 13$ sites.

4.2. Curie temperatures

Saturation magnetization versus temperature analyses for eight basalt specimens from both Yingan and Camp 66 indicated a general consistency in magnetic behavior (see Fig. 8). Curie temperatures span a range from 400° to 550°C. Regardless of the specific nature of the curves, no directional dependence is observed suggesting that the reversed magnetizations are relatively unaffected by chemical alteration.

4.3. Fold test

An opportunity for the application of Graham's [22] fold test exists between the three sections (see Fig. 9). If directions of magnetization are brought into agreement after bedding planes are restored to horizontal, the magnetizations have remained unchanged since the time of deformation. Using McElhinny's [23] statistical procedure to evaluate overall improvement in the grouping of the reversed directions, fold tests for Subasi and Yingan ($k_2/k_1 = 6.36$, $n = 19$ sites), and Subasi and Camp 66 ($k_2/k_1 = 9.35$, $n = 24$ sites) are both significant at the 95% confidence level. Assuming that all three directions originally belonged to a single

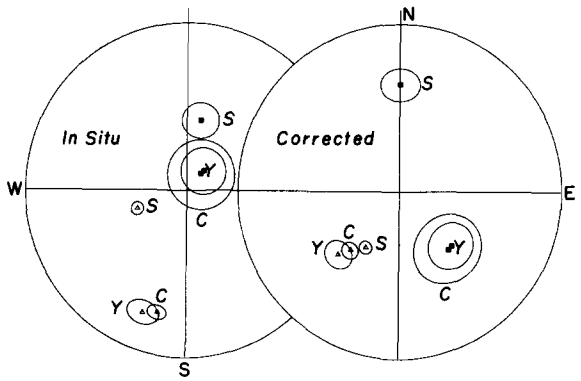


Fig. 9. Equal area stereographic projections of reversed primary section mean directions and normal secondary section mean directions each with circles of 95% confidence both before (left) and after (right) correction for tilt of beds. Subasi section means labeled with an *S*, Yingan means labeled with a *Y* and Camp 66 means labeled with a *C*. Primary directions (open symbols, upper hemisphere) clearly converge with correction while secondary directions (closed symbols, lower hemisphere) diverge.

TABLE 1 (continued)

N = number of samples; D = in situ declination in degrees; I = in situ inclination in degrees; D_c = corrected declination in degrees; I_c = corrected inclination in degrees; R = resultant vector; k = precision parameter estimate; α_{95} = radius of 95% confidence in degrees; VGP Lat. N = virtual geomagnetic pole latitude in degrees north; VGP Long. E = virtual geomagnetic pole longitude in degrees east. The mean direction is a Fisherian average of site mean directions. The pole position is obtained from a Fisherian average of VGP positions.

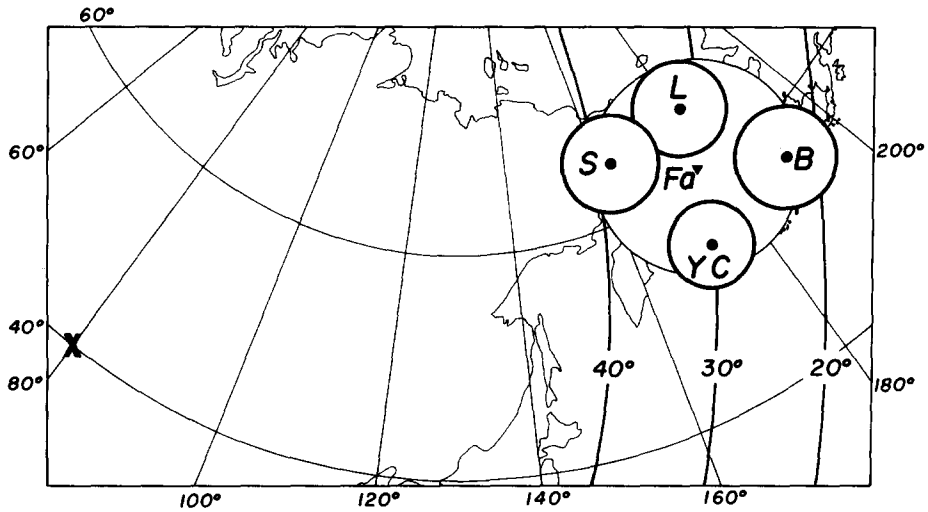


Fig. 10. Lambert Conformal Conic map with principal latitudes of 60 ° N and 35 ° N, and a central meridian of 130 ° E of Permian data from Tarim. Pole positions are plotted as follows: Bai et al.'s [11] pole labeled *B*, Li et al.'s [12] dike pole labeled *L*, Subasi limestone pole (this study) labeled *S* and the Yingan–Camp 66 basalt and limestone pole (this study) labeled *YC*. Position labeled *Fa* is the Fisherian average of the four Permian poles. Twenty degree, 30 ° and 40 ° lines of paleolatitude about the sampling site (40 ° N, 80 ° E; marked with an *X*) are also drawn.

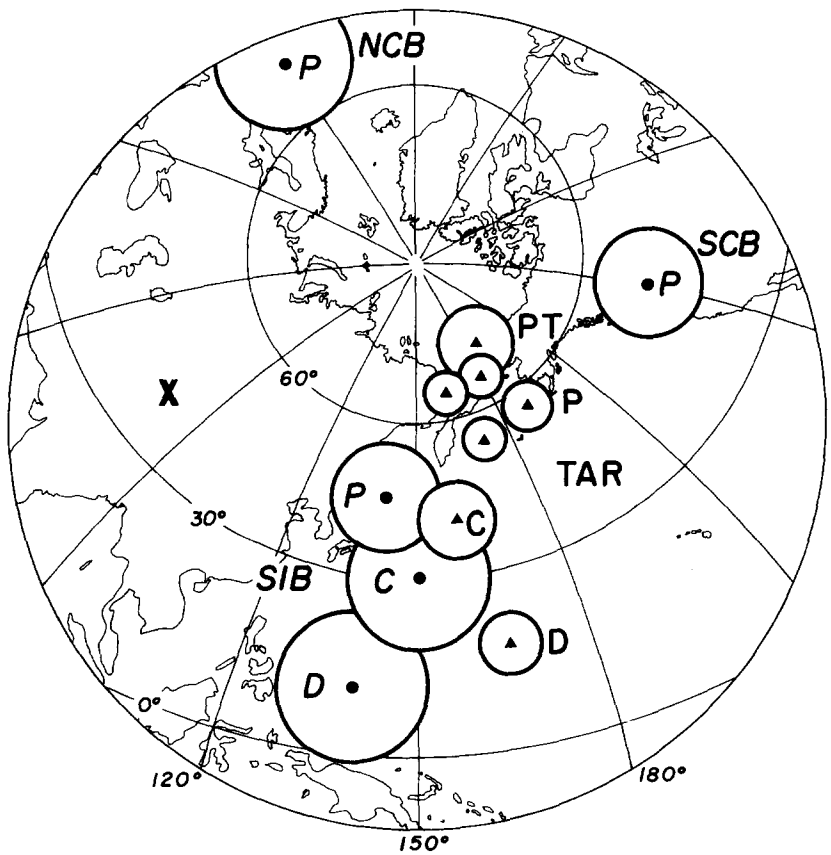


Fig. 11. Stereographic projection, centered about the point 60 ° N, 150 ° E, of Paleozoic data from Tarim (triangles) and Siberia (circles) [26], and Permian data from North and South China (circles) [9]. Data from Tarim include a Devonian pole [14], an Upper Carboniferous pole [10], a Lower Permian pole [11], the Yingan–Camp 66 basalt and limestone pole (this study), the Subasi limestone pole (this study), an Upper Permian pole from dikes [12] and a Permo-Triassic pole [15]. Sampling area for Tarim data marked with an *X*.

population, the statistical argument of McFadden and Jones [24] indicates that the limestone mean direction at Subasi after tectonic correction is still distinguishable from the corrected mean directions at Yingan and Camp 66. However, the hypothesis of a common true mean direction between the means at Yingan and Camp 66 cannot be rejected at the 95% confidence level, and thus we have grouped these two populations. The Subasi limestone pole and the combined Yingan–Camp 66 pole ($\lambda = 54.5°$ N, $\phi = 172.3°$ E, $K = 66$, $A_{95} = 4.0°$, $n = 21$ sites) differ primarily in colatitude ($8.1 \pm 6.5°$) suggesting a possible difference in magnetization age (see Fig. 10).

The normal components of magnetization clearly diverge with structural correction, suggesting that these represent an overprint acquired after tilt of beds (see Fig. 9).

5. Tectonic interpretation

A comparison of our results with those of Zhai et al. [10], Bai et al. [11], Li et al. [12], Li [14], and McFadden et al. [15] for Tarim during the Permian and earlier Paleozoic indicate continuing apparent polar wander, from the Devonian through the Carboniferous and Permian to the Permo-Triassic boundary (see Figs. 10 and 11). The three Lower Permian pole positions differ, and the differences correlate well with changes in lithologies and locations of sampled sections. Three explanations could account for such differences. First, the three positions could have different magnetization ages. The Subasi limestone pole is closest to Tarim, suggesting steeper inclinations recorded at higher paleolatitudes. The results of Bai et al. [11] yield the shallowest inclinations. The section sampled at Camp 66, however, is identical to the Sishichang section sampled by Bai et al. [11], but separated by only 50 km in strike distance. These two pole positions should agree (see Fig. 10).

Second is the possibility of a depositional inclination error [21]. Bai et al.'s pole appears to be most susceptible to an artificial flattening of inclination. Of the 26 sites used in the calculation of their pole position, ten were collected from red sandstones and another ten from quartzose or yellow-green sandstones. These yielded a mean inclination ($\bar{I} = -41.0°$) which is 10° shallower than our Camp 66 mean inclination ($\bar{I} = -51.4°$), reminiscent of the difference observed in this paper between the Kalundal sandstones and the Balikelike limestones at Subasi. A depositional inclination error could affect the magnetization of the Balikelike limestones. This, however, would make an already steep inclination steeper, thus increasing the initial amount of discordance between the Subasi pole and the other two (see Fig. 10).

Third, discrepancies between all three Lower Permian poles and the Upper Permian pole could be a consequence of small relative motions between faultbounded blocks along Tarim's northwestern margin. The discordance between our Yingan–Camp 66 basalt and limestone pole and the result from Upper Permian basaltic dikes found also at Yingan but principally at Bachu some 100 km to the southwest [12] is almost entirely in terms of declination (see Fig. 10). Local rotations about a near-vertical axis for either section could easily reconcile this difference, suggesting that the dikes and flows are related in age. The possibility of local rotations limits the interpretation of directions of magnetization from individual tectonic units.

Relative rotations and inclination error explain differences observed between pairs of the four poles as previously discussed, yet all discordances could simply be a consequence of magnetization age. Our strategy has been to average inclinations from the four units to produce a Permian paleomagnetic inclination applicable to the underlying block. A Fisherian average of the four ($\lambda = 60.9°$ N, $\phi = 177.0°$ E, $K = 98$, $A_{95} = 9.4°$, $n = 4$ poles) produces a conservative Permian pole position for Tarim (see Fig. 10). Inclinations predicted by the Fisherian average are equivalent to inclinations from an inclination-only average [25]. Predicted declinations may be biased though, due to the possibility of a systematic rotation of the entire northwestern rim of Tarim. Additional Paleozoic results from other areas are needed, specifically from the southern margin of Tarim (see Fig. 1).

Permian mean paleomagnetic poles from Tarim, Siberia [26], North China and South China [9] all differ significantly, suggesting substantial post-Permian relative motions (see Fig. 11). The latitudinal proximity of Tarim to Siberia during the Permian supports the claim that Tarim, the most

inboard of the Chinese block, arrived first. Upper Permian flora and fauna distinct from those of North and South China, but identical to Angaran species lends additional support to a time of accretion of Tarim to Siberia before either North or South China [17].

Similarities in the shapes of the Late Paleozoic apparent polar wander paths for both Tarim and Siberia argue for pre-Triassic motions about similar Euler poles (see Fig. 11). Tarim could have ridden passively along on a lithospheric plate including Angaraland, however, substantial relative motions have occurred since the Paleozoic, as the collision and subsequent sinistral strike-slip motion of Tarim with respect to a stable Siberia are well documented in the paleomagnetic and stratigraphic records [27].

6. Conclusion

Characteristic magnetizations obtained from the three stratigraphic sections are probably original Lower Permian directions for four reasons. First, the reversed magnetizations are consistent with the Permo-Carboniferous Kiaman Reversed Interval [28]. Second, varying lithologies and types of remanent magnetization give consistent directions at widely spaced localities. Third, the characteristic directions pass a fold test at the 95% confidence interval, establishing the Early Tertiary as an upper boundary on magnetization age. Finally, apparent polar wander and polarity changes exist in Devonian sediments exposed in the continuous section at Subasi, ruling out the possibility of a post-Permian regional overprint.

Small (11°) discordances between the Subasi limestone paleomagnetic pole, the Yingan–Camp 66 basalt and limestone pole, the Lower Permian pole of Bai et al. [11] and the Upper Permian pole from basaltic dikes reported by Li et al. [12] suggest either differences in magnetization age, inclination error or relative rotations between individual sampled tectonic units. A combination of all three is most plausible and best explains the discordant nature of the four Permian pole positions. A Fisherian average of the four poles is conservative and produces a reference inclination for Tarim during the Permian. Declinations could be biased due to the possibility of a systematic rotation of the entire northwestern rim of Tarim with respect to the stable interior. Differences in excess of biases introduced by any potential local rotations exist between Permian paleomagnetic poles from Tarim, Siberia, North China and South China suggesting substantial post-Permian relative motions.

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